SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 5 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I **Subash G (192572150)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Annexure A

Short Answer Test

Question 1 (5 Marks)
Suppose a modern Operating System supports multiple users and processes with shared resources like CPU, memory, and I/O devices.
(a) Classify different types of Operating Systems suitable for this scenario and justify your choice.
(b) Draw the structure of an Operating System showing how resources are managed.
(c) Write a sequence of system calls required to create and execute a new process in such an environment. **(BL2)**

Question 2 (5 Marks)
Consider an Operating System managing multiple files and devices for concurrent users.
(a) Categorize system calls used for file and device management.
(b) Illustrate how the OS handles file operations using a layered structure diagram.
(c) Write a pseudo-code or C-based system call sequence to open, read, and close a file. **(BL3)**

Question 3 (5 Marks)
Assume a real-time Operating System used in an embedded system controlling industrial machinery.
(a) Identify the type of OS and explain why it is suitable for real-time applications.
(b) Draw a diagram representing the interaction between hardware, kernel, and user processes.
(c) List and explain system calls required for process synchronization in this system. **(BL2)**

Question 4 (5 Marks)
A distributed computing system consists of multiple interconnected computers sharing resources over a network.
(a) Classify the type of Operating System used and compare it with centralized OS.
(b) Draw the architecture of a distributed OS showing communication between nodes.
(c) Write the sequence of system calls involved in inter-process communication (IPC). **(BL3)**

Question 5 (5 Marks)
Consider a scenario where an Operating System must manage process scheduling, memory allocation, and device communication efficiently.
(a) Explain the major functions of an OS in handling these tasks.
(b) Draw a block diagram representing OS components and their interactions.

(c) Write system call examples for memory allocation and device management. **(BL3)**

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

COI - AT1.

SHORT ANSWER TEST

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Course code - CSA0409

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(1) Different types of operating

1. Multi-user OS

Allows multiple users to use the same computer/ system simultaneously.

Example: Linux, UNIX Etc...

2. Multitasking OS;

Allows multiple processes to Execute concurrently by sharing CPU time.

Example: Linux, windows, macOS.

In this many processes may be running at the same time, so the OS must schedule CPU time among them.

3. Multiprocessing OS

Supports multiple CPUs / cores working together.

Example: windows System.

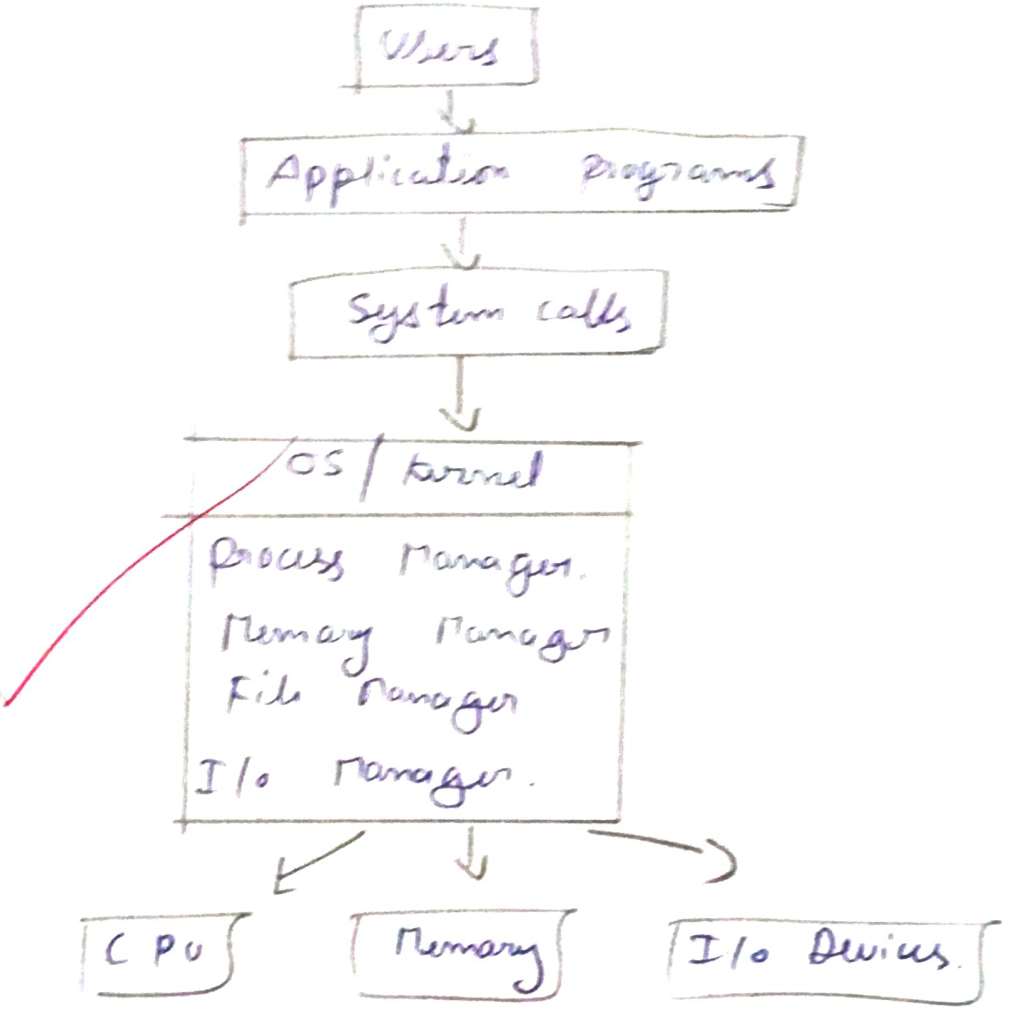
Justification: CPU Resources and modern System commonly use multiple processor cores to improve performance.

4. Time - Sharing OS

CPU time is divided into small time slices and allocated among user / processes

It provides Responsive Interaction when several users processes are competing for CPU resources

b) Structure of an operating System showing resources are managed;



c) System calls to create & execute a new process

fork()
↓
exec()

wait()
↓
exit()

Example:

```
pid = fork();
exec(...);
wait(Null);
exit(0);
```

---

2) Categaring system calls for devices management.

a)

I. File Management calls;

* Open()
* Read()
* write()
* Iseek()
* close()

Example:

```
fd = open("data.txt", ...);
read(fd, buffer, ...);
write(fd, buffer, ...);
close(fd);
```

2. Device Management calls.

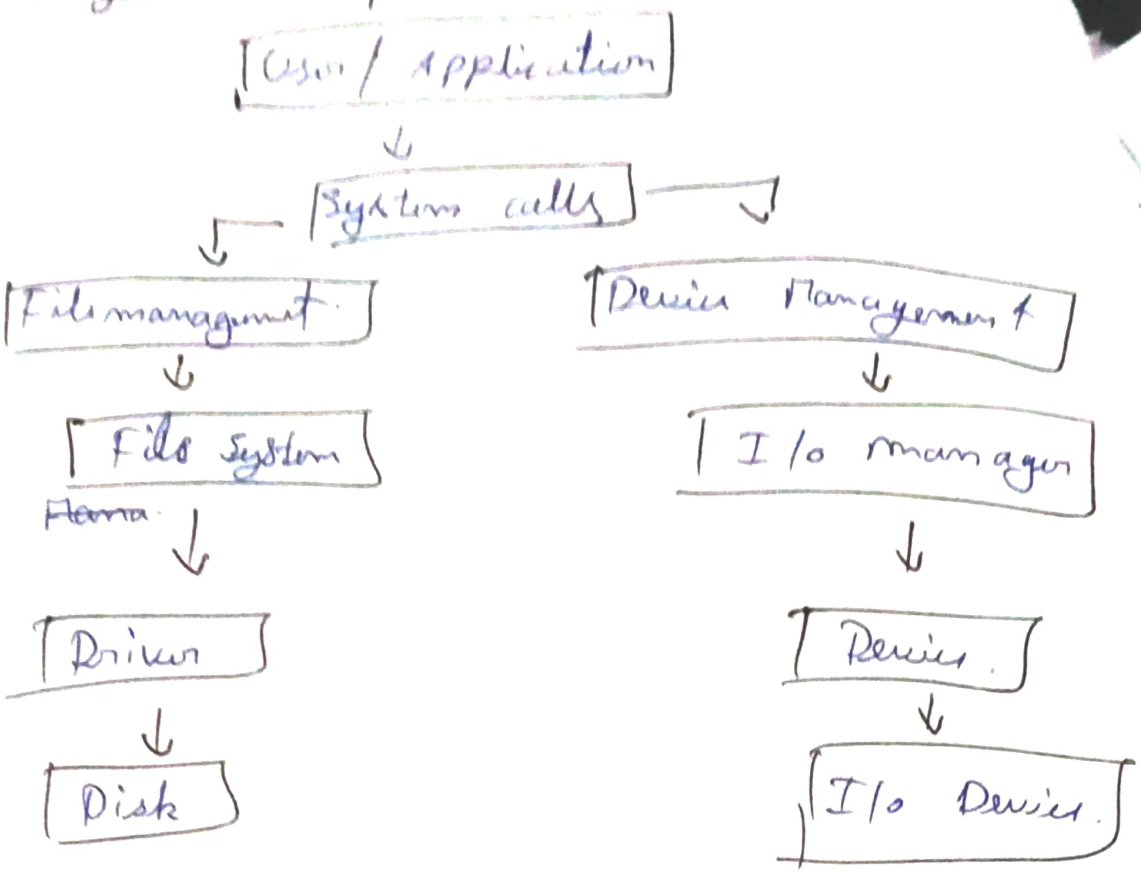
Used to communicate with and control I/O devices

* Read()
* ioctl()
* Open()
* close()

Example:

Program
↓
read()
↓
OS - Kernel
↓
Device Driver
↓
Keyboard / Disk / printer

(b) OS manages the operations:



c) System call sequence to open, read & close a file;

```
int main()
{
    int fd;
    char buffer[100];
    fd = open("data.txt", O_RDONLY);
    read(fd, buffer, sizeof(buffer));
    close(fd)
}
```

a) OS type & Explanation ;

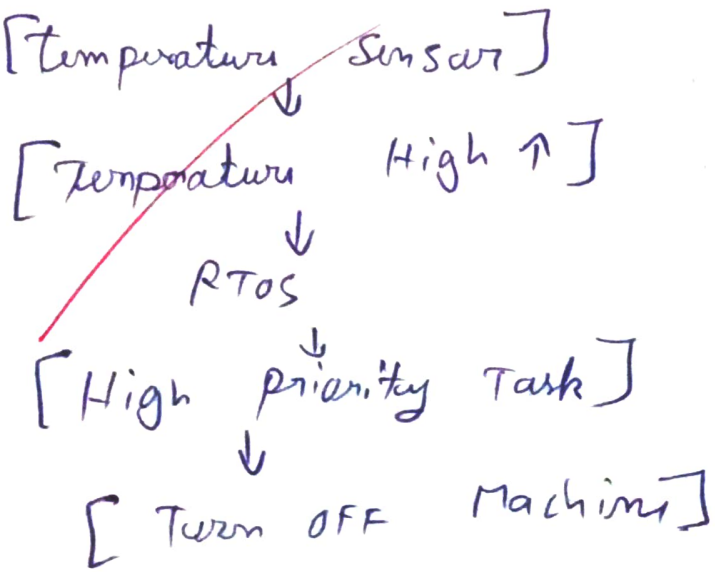
The OS type will be " Real time operating System" (RTOS)

An RTOS is an operating System disigned to Respand to event within a Specified time limit (deadline).

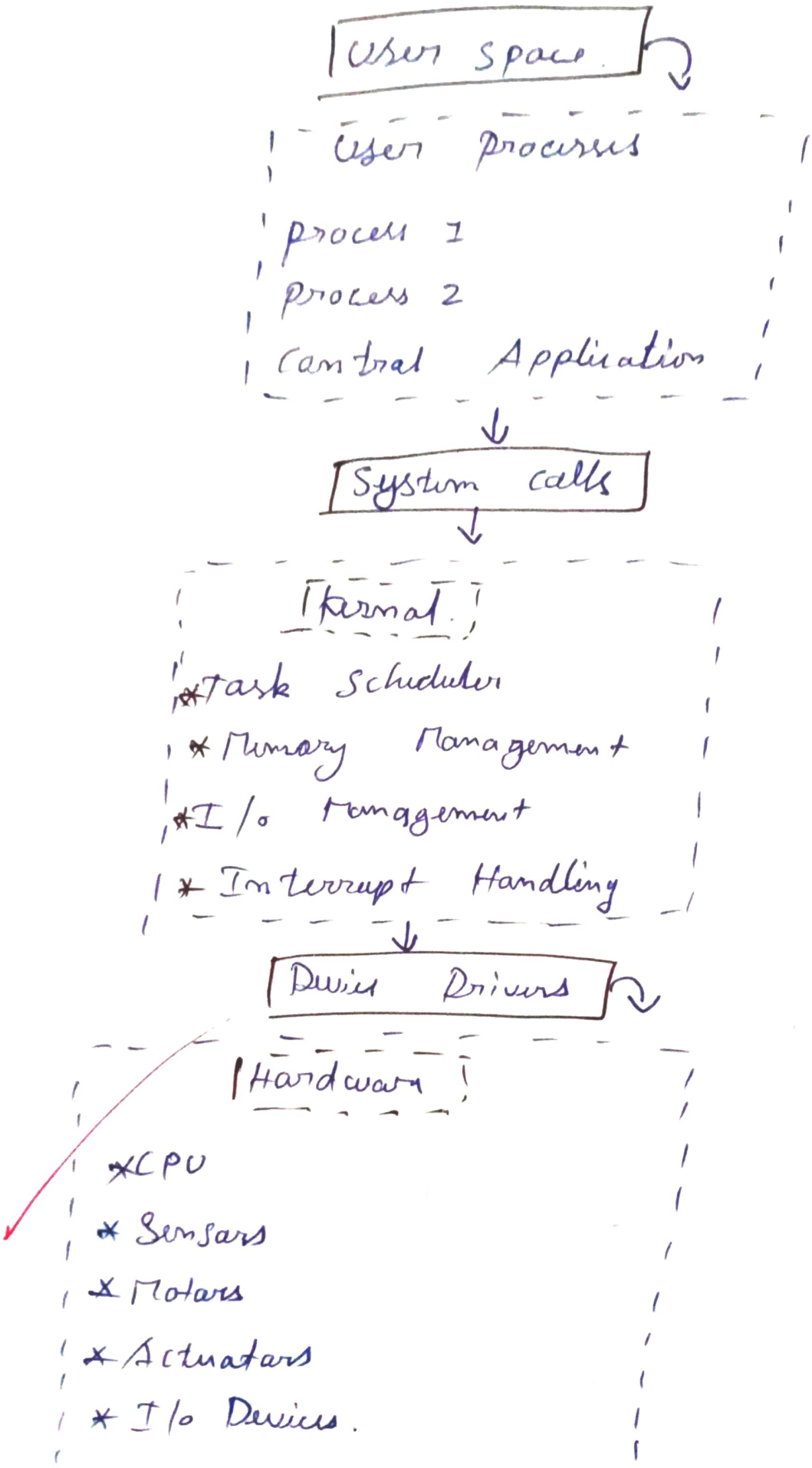
The RTOS is Suitable Because:

1.) predictable Responses
2) priority Based Scheduling
3) Fast Respance to interrupts
4.) Reliable operation
5) Deadline - based execution.

Example:



b) Interaction btw hardware, kernel & user



System calls Required for Synchronization,

1. Sem_wait()

Used to wait for / acquire a Semaphore.

Sem_wait()
↓
Check Resource
↓
Resource available?
↓
Acquire it

2. Sem_post()

Used to Release / ~~sig~~ signal a Semaphore.

Task finishes
↓
Sem_post()
↓
Resource Released.

3. Mutex operations;

It provides mutual Exclusion;

Lock()
↓
Critical Section
↓
unlock

* Only one task can enter the critical Section at a time

4)

(a) classify the type of operating System:

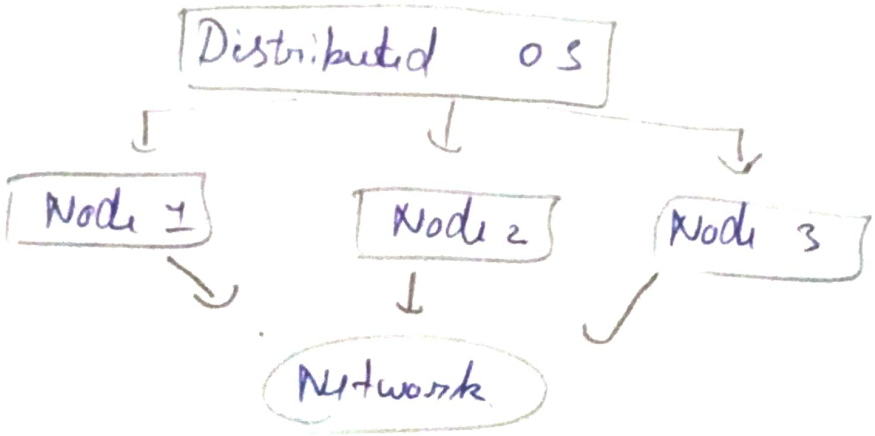
Ans to given situation;

The System Contains:

[multiple nodes + network + resource Sharing]

The Best OS : Distributed operating System.

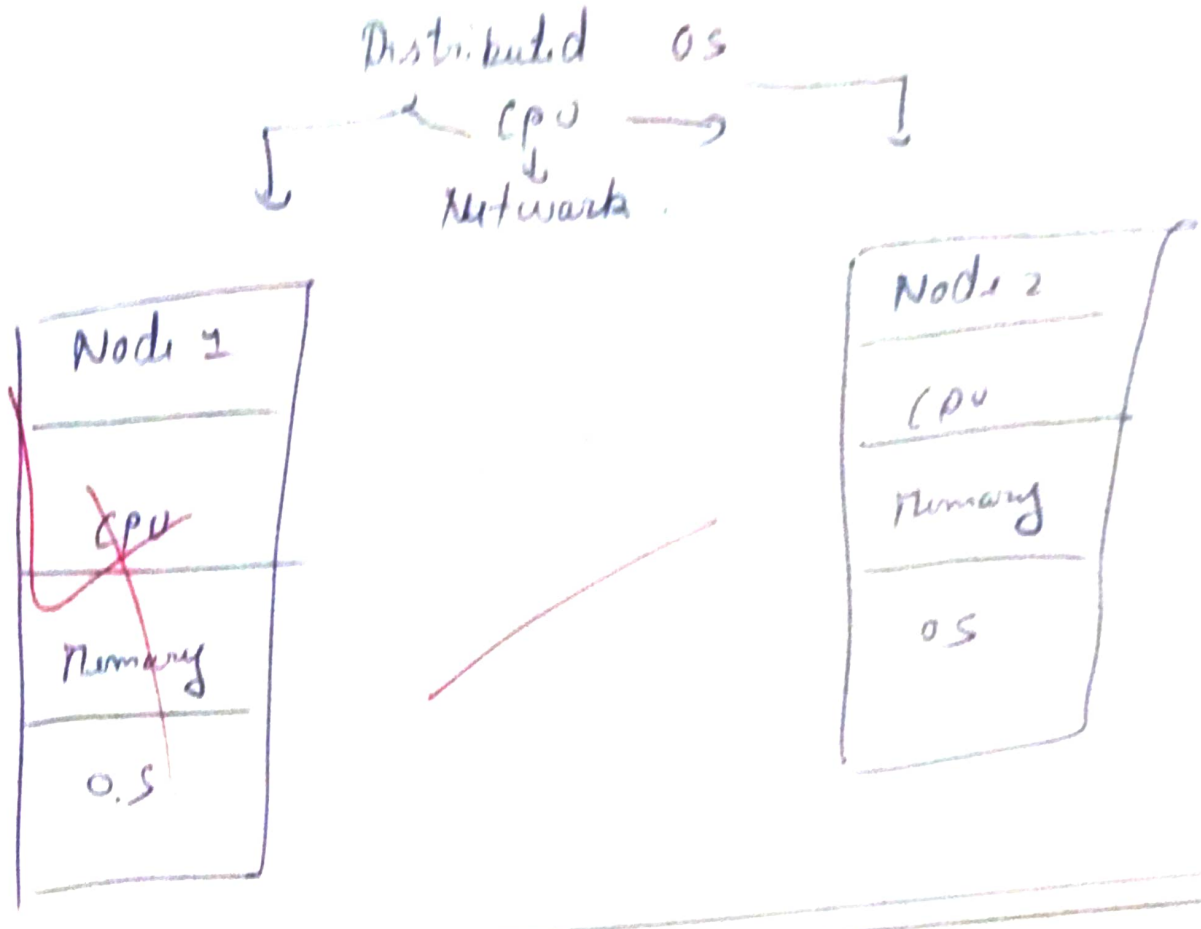
It manages a collection of independant computers and makes them work together as a coordinated system.



It has;

* multiple computers
* Network connectivity
* Resource Sharing
* Distributed Processing.
* Fault tolerance.

Architecture of Distributed OS;



5.)

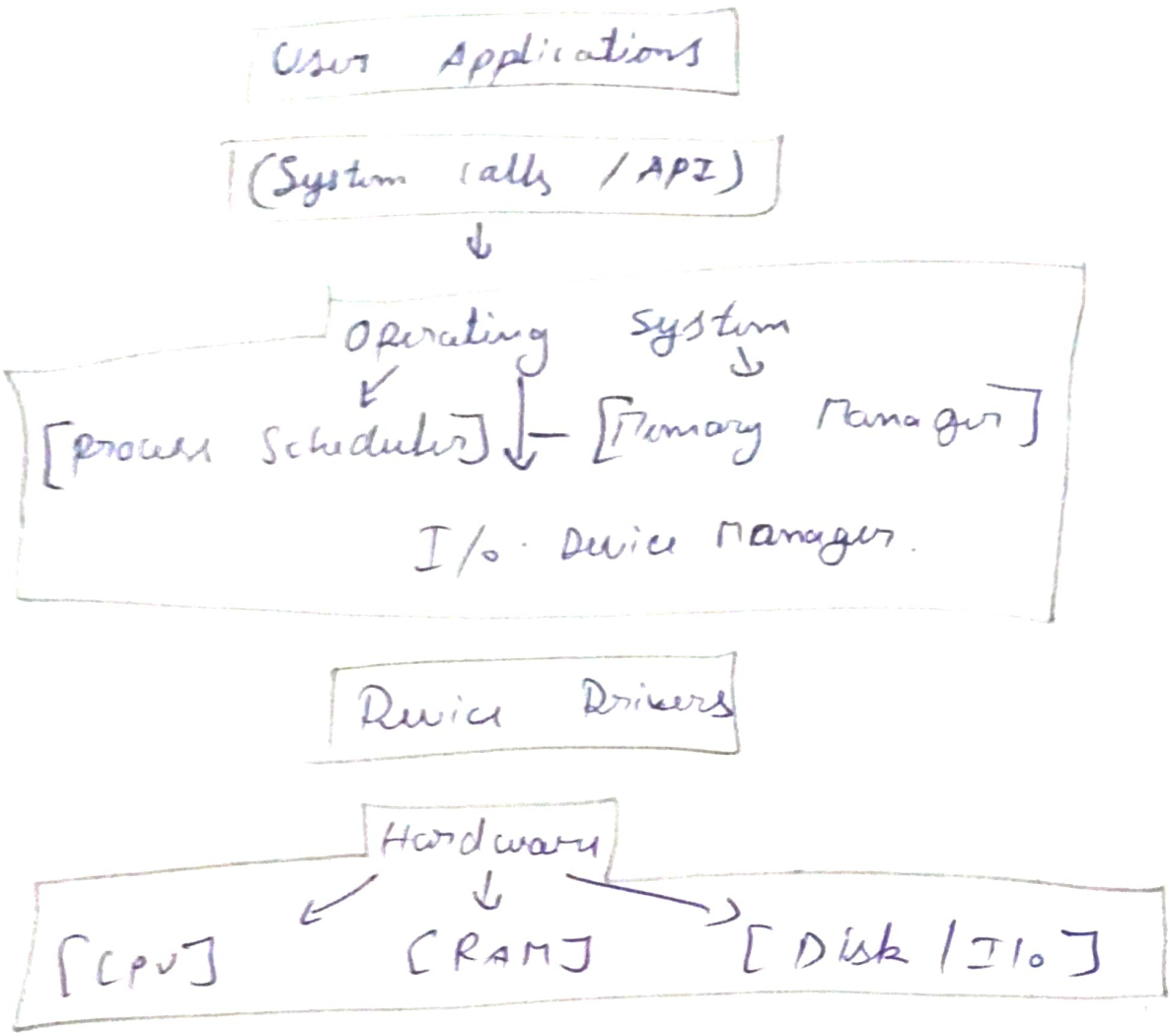
(a) Major Functions of an OS in Handling tasks;

(i) process Scheduling - the OS tracks the status of all the processes and determines which process receives the CPU, when and for how long.

(ii) Memory Allocation: The OS manages the primary memory (RAM). It keeps track of which parts of memory are currently using.

(iii) Device communication; the OS acts as an intermediary between hardware devices and programs

(b) Block diagram of OS components :



(c) System call Examples :

Memory Allocations ;

* brk() / sbrk() → changes the space allocated for the calling process's data segment.

* VirtualAlloc() — Reserves, commits or changes the state of a Region of pages in in virtual Address Space.

Management:

- open() → opens a driver file for reading / writing
- ioctl() → sends driver - specific parameters & operations

* CreateFile() → Create a driver handle
* DeviceIoControl() → Sends a control code

